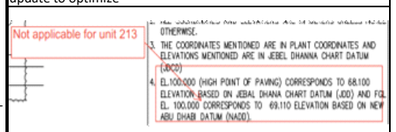
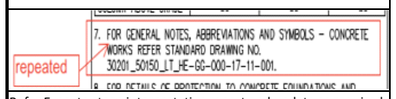
wood.	ADNOC GAS LONG TERM - EXECUTE STAGE PROJECT 5.0 (P5) EPCm FOR HABSHAN DBN COMPANY PROJECT No. 30201-50150-LT-HE WORK ORDER NO. 3000076501

Comments Resolution Sheet (CRS) - EPCm

COMPANY Trans. No.:	ADNOCGP-WTRAN-023937	Received Date:	2-Feb-26
COMPANY Doc. No.:	30201_50150_LT_HE-H4-280-17-23-001	Revision:	B
Document Title:	UNIT 280 -HABSHAN 4 FOUNDATIONGA & DETAILS SUBSTATION SS-201	Status Code:	2
ADNOC PM:	Dr Azharuddin Mohammed	EPCm. CONTRACTOR PM:	Shailesh
ADNOC Reviewer:	Mr Vinodhkumar Ranganathan	EPCm. CONTRACTOR Originator:	Hamed Gholampour

Review Codes: ☐ Code-1: Rejected ☑ Code 2: Comments as Noted ☐ Code-3: No Comments ☐ Code 4: For Information Only

REVISION B

No.	COMMENT SECTION/ CLAUSE / DRAWING REFERENCE	COMPANY Comment	COMPANY Reviewer	EPCm CONTRACTOR Response	Open/Closed	COMPANY Response (If Comments are still open / Not incorporated or if Eng Contractor Response Not Acceptable)	EPCm CONTRACTOR Response	Open/Closed	Remarks
1	30201_50150_LT_HE-H4-280-17-23-001-01	Refer 30201_50150_LT_HE-H4-280-17-23-003 and update for the typical comments and presentation	Mr Vinodhkumar Ranganathan	Noted and Updated					
2	30201_50150_LT_HE-H4-280-17-23-001-02	Check the correctness of type and update		Noted and Updated					
3		For cellar room , foundation and column size may be optimized. check and update. Typical comment		Noted. SBC very low as per the latest geotechnical report, optimization done as much poosible.					
4		Check the correctness and update		Noted and Updated					
5		Same foundation size considered in all the columns. check and update to optimize		Noted and Updated					
6				Noted and Updated					
7				Updated					
8		Refer Execute stage interpretative report and update as required. There is a change in SBC.		Noted. Foundations updated for the SBC, as per the new geortechnical report.					
9		check to optimize and update as required		Noted and Updated					
10		check whether this is applicable for this area and update		Noted and Updated					
11		Refer SS#203 drawing and update in same manner		Noted and Updated					
12	30201_50150_LT_HE-H4-280-17-23-001-03	Update to show the Joint details to execute together with concrete casting		Noted and Updated					
13		For combined foundation, top rebar in critical. check and update		Noted and Updated					
14		update to provide the rebar plan		Noted and Updated					
15		T12 may be sufficient for side face rebars		Noted and Updated					
16		As discussed, ensure to provide side face rebar at a spacing < 300mm.		Noted and Updated					
17	30201_50150_LT_HE-H4-280-17-23-001-04	SS-203 span is more or less same as SS-201. But rebar detailing is different. Check and ensure uniformity in design is maintained. update		Noted and Updated					
18	30201_50150_LT_HE-H4-280-17-23-001-05	consider cap u-bar		Noted and Updated					
19		Heavy dia. may not be required for distribution. check and update		Noted and Updated					
20		Update the Fire wall section and details same like SS-203.		Noted and Updated					
		As discussed with EPCM Architect, prepare the study sketch demonstrating the height requirement of fire wall to discuss with CPY TEs before the next submission of this drawing.		As Discussed with company, this detail to reamain as it is.					
		Please refer the E08 type standard earthing installation from drawing 30201_50150_LT_HE-GG-000-16-21-001 for REINFORCEMENT CONCRETE STRUCTURE	Mr Kiranbhai kuntmal	Noted and Updated					

No.	COMMENT SECTION/ CLAUSE / DRAWING REFERENCE	COMPANY Comment	COMPANY Reviewer	EPCm CONTRACTOR Response	Open/Closed	COMPANY Response (If Comments are still open / Not incorporated or if Eng Contractor Response Not Acceptable)	EPCm CONTRACTOR Response	Open/Closed	Remarks
		The curved cover shall be extended to the wall and the height of the fire wall between the transformers shall not be extended to the roof to allow for air circulation. Refer to the discussion during meeting of building design.	Mr Mohamed Elsadani	As Discussed with company, this detail to reamain as it is.					
REVISION A									
1	30201_50150_LT_HE-H4-280-17-23-001-01	As already commented, ensure to finalize the Architectural drawings and align with all the respective disciplines. Highly recommend to arrange a session/workshop to resolve the Hold list on Archi. drawings and agree on the building size, room size, building location and cable entry locations & levels before the next submission.	Mr Vinodhkumar Ranganathan	Hold list shall be removed after approved Architectural drawings and other discipline drawings.					
2		EPCM to provide the status of other disciplines critical deliverables and demonstrate the basis of issuing the Foundation drawings for construction.	Mr Vinodhkumar Ranganathan	Noted					
3		Refer the comments in design report and update as required.	Mr Vinodhkumar Ranganathan	Comments on design report has not been received yet.					
4		Any update required based on the Final plot plan shall be carried out.	Mr Vinodhkumar Ranganathan	Comments on design report has not been received yet.					
5		Any update required based on the model review comments shall be carried out.	Mr Vinodhkumar Ranganathan	Noted.					
6		Any update required based on the Execute stage Geotech reports shall be carried out.	Mr Vinodhkumar Ranganathan	Noted.					
7		Any update required based on the CPY HSE approved QRA/FERA report shall be carried out.	Mr Vinodhkumar Ranganathan	Noted.					
8		Update the status of CP vendor. Ensure to align with CP vendor and provide the CP installation drawings to GCC1 to execute under the supervision of CP vendor. Expedite to conclude at the earliest.	Mr Vinodhkumar Ranganathan	Noted & incorporated					
9		Finalize the cable entries/exit and ensure the clash free and feasible proposal	Mr Vinodhkumar Ranganathan	Cable trench routing is hold and will be checked after finalizing the routs.					
10		Add section to show reinforcements for wall	Mr Mandar Nadgouda	The superstructure is not non-blast and RC wall has been removed.					
11		Update to show elevation indicating opening size and location. Also show reinforcement details including details for starter bars/ dowels details around openings. Typical comment check and update at all wall opening.	Mr Mandar Nadgouda	All openings will be shown at structural drawings after finalization by other disciplines					
12		Update to add detail showing reinforcement at junction of wall and column.	Mr Mandar Nadgouda	The superstructure is not non-blast and RC wall has been removed.					
13		Design summary / calculation of individual or combined foundation not included in the design calculation. Hence could not be verified. Ensure to align drawings with calculation. Also ensure to include all the native files in design calculation report.	Mr Mandar Nadgouda	Noted & incorporated					
14		Confirm that the Transformer yard size and spacing approved by CPY Electrical and CPY HSE.	Mr Vinodhkumar Ranganathan	Noted & incorporated					

No.	COMMENT SECTION/ CLAUSE / DRAWING REFERENCE	COMPANY Comment	COMPANY Reviewer	EPCm CONTRACTOR Response	Open/Closed	COMPANY Response (If Comments are still open / Not incorporated or if Eng Contractor Response Not Acceptable)	EPCm CONTRACTOR Response	Open/Closed	Remarks
15		Review for optimizing the fire walls by combining into one and update	Mr Sreekanth Nuthanapati	Noed. Will be incorporated after final architectural drawings received.					
16		It is presumed that these trenches are above foundation. Please check and confirm. Also show it in cross-section for clarity	Mr Abhishek Srivastava	Noted. Can be confirmed after receiving final trench layout.					
17		This is not in archi. drawings. check and align each other. Update	Mr Vinodhkumar Ranganathan	Architectural drawings will be updated.					
18		Add note for GWT to mention in terms of elevation.	Mr Mandar Nadgouda	Noted & incorporated					
19		Holds to be resolved before next revision.	Mr Mandar Nadgouda	Noted.					
20		Recommend to provide in tabular format similar to unit 277 PRK Fdn drawings. Check and update as necessary	Mr Mandar Nadgouda	Noted & revised					
21		Align the staircase along the building. Check and update	Mr Vinodhkumar Ranganathan	Staircases on hold till final architectural drawings received.					
22		check to optimize extension	Mr Mandar Nadgouda	Noted & incorporated					
23		Provide distance from grid. Typ. comment.	Mr Mandar Nadgouda	Noted & incorporated					
24		check and update	Mr Mandar Nadgouda	Noted & incorporated					
25		Update to include sealant details as required.	Mr Mandar Nadgouda	Will be updated in next revision.					
26		Review and update to include filler board as it is not possible to place / remove shuttering post concreting - typical comment	Mr Sreekanth Nuthanapati	Will be updated in next revision.					
27		Recommend to bifurcate foundation in appropriate zones of higher shear / bending and check to optimize reinforcement in the zones with relatively lower stress. Also provide plan to show rebar layout with splicing at suitable locations Update.	Mr Mandar Nadgouda	Building changed fron blast to non-blast and strip footing has been removed.					
28	30201_50150_LT_HE-H4-280-17-23-001-02	Check and update for Side bars for all footings.	Mr Mandar Nadgouda	Noted & incorporated					
29		For strip foundation, Rebar plan layout and sections are mandatory showing the curtailments, lapping distance, locations etc. The shown rebar detail is very conservative. Update to provide the rebar details only as required with the suitable contingencies.	Mr Vinodhkumar Ranganathan	Building changed fron blast to non-blast and strip footing has been removed.					
30		Ensure Ldc for column main longitudinal rebar is satisfied as per footing thickness provided, Justify with calculation. Check and update as necessary Typical comment for all footings.	Mr Mandar Nadgouda	Noted & Incorporated.					
31	30201_50150_LT_HE-H4-280-17-23-001-03	Reinforcement details shown are not matching with calculation report. Check and ensure to align with design calculation. Typical comment for all columns	Mr Mandar Nadgouda	Noted & incorporated					
32		Show near trench also to clarify that ir is clearing foundation (typ.)	Mr Abhishek Srivastava	Will be shown after finalization of trenches location.					
1		Horizontal rebars shall be on the external. Update	Mr Vinodhkumar Ranganathan	Noted & Incorporated					
2		add dimension to locate or add suitable note for locating rebar laps and staggering of splice	Mr Mandar Nadgouda	Reffer to note no.6					
3		Show side bars	Mr Mandar Nadgouda	No need to Side bar for this foundation.					
4	30201_50150_LT_HE-H4-280-17-23-001-04	Update to show the grooves	Mr Vinodhkumar Ranganathan	Will be incorporated after final architectural drawing received.					
5		This design is good for heat circulation. However, falling rain might leak on top of Transformer cable box. Hence, change as shown with slop.	Mohamed Elsadani	Will be incorporated after final architectural drawing received.					
6		Ensure to extend beyond edge to ensure that rain water will drain on the sheeting, update applicable drawings	Mr Sreekanth Nuthanapati	Will be incorporated after final architectural drawing received.					
7		This design is not acceptable. why the fire wall is so high. Revise as disscussed	Mr Abhishek Srivastava	shall be incorporated in next revision.					

DRAWING NO.	DRAWING TITLE	SHEET	SHEET TITLE
30201_50150_LT_HE-H4-280-17-23-001	UNIT 280 – HABSHAN 4 FOUNDATION GA & DETAILS SUBSTATION SS-201	01	DRAWING INDEX
		02	BUILDING FOUNDATION LAYOUT
		03	FOUNDATION & COLUMN RC DETAILS
		04	COLUMN SECTION & DETAILS
		05	WALL SECTION & DETAILS

SCALE 1:10
0 0.2 0.4 0.6 0.8 1.0m

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
0	07-FEB-2026	ISSUED FOR CONSTRUCTION WITH HOLD (IFC-H)	RM	DR/KS	VV
B	30-JAN-2026	ISSUED FOR APPROVAL	KR	HG/KS	VV
A	17-JAN-2026	ISSUED FOR COMPANY REVIEW (ICR)	VA	AC/HB	VV/MS

SCALE:	DRAWN:	CHKD:	APPRD:
1:10	ASHOK	ANKITA/BALAJI	VISHWANATH/SOFI

أدنوك للغاز
ADNOC Gas
ADNOC

EPCm CONTRACTOR: wood.
EPCm CONTRACTOR: AMEC INTERNATIONAL LTD.

PROJECT TITLE
PROJECT 5.0 (P5)
LONG TERM - EXECUTE STAGE
EPCm FOR HABSHAN DBN

DRAWING TITLE
UNIT 280 –HABSHAN 4
FOUNDATION GA & DETAILS SUBSTATION SS-201
DRAWING INDEX

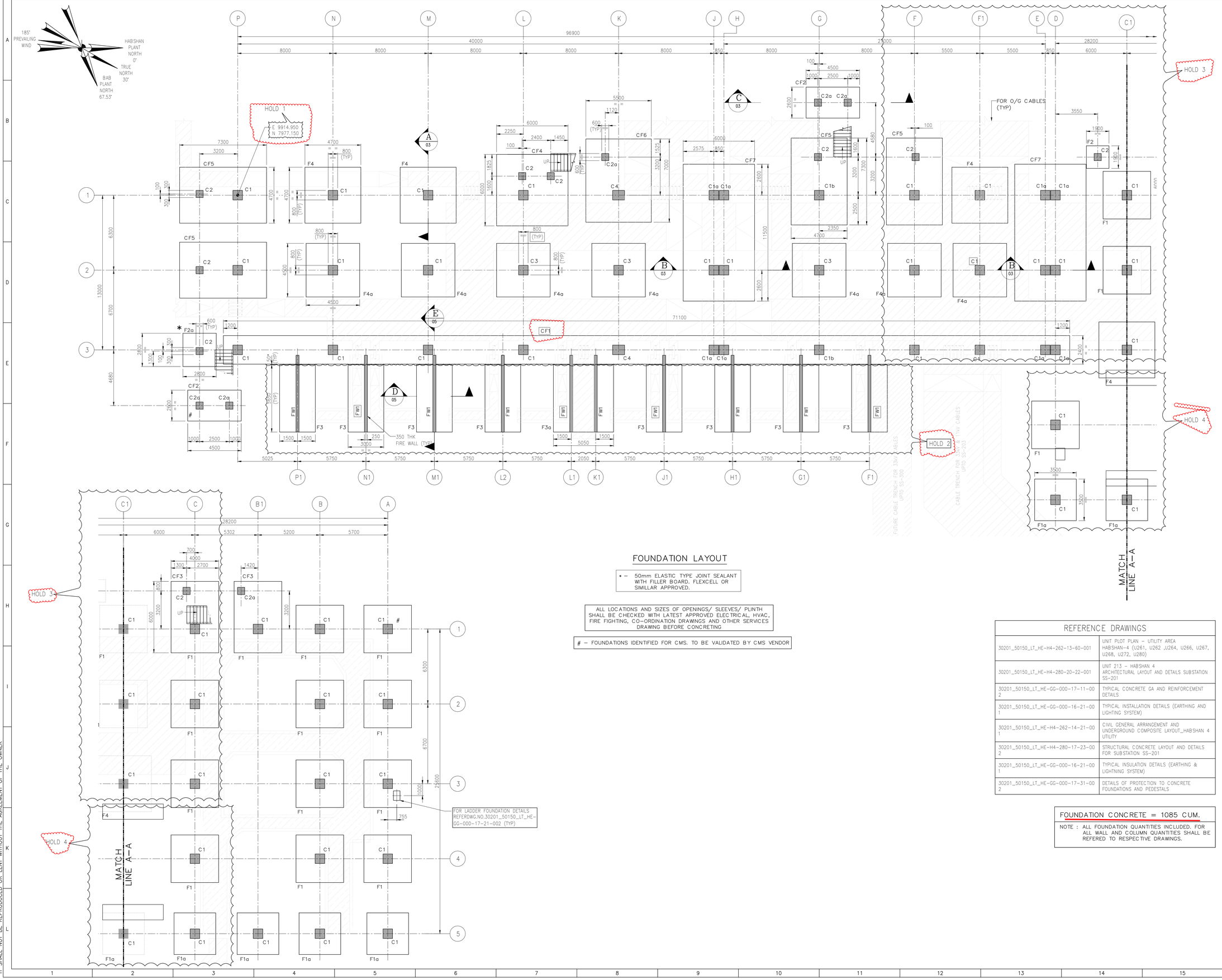
UNIT: 213 PLANT: H4
CONTR. PROJ. No. 3000076501 CONT. DWG. No.: XXXXXXXX
ADNOC GAS CONTRACT No. 3000076501
ADNOC GAS PROJ. No.: 30201_50150_LT_HE ADNOC GAS DWG. No.: 30201_50150_LT_HE-H4-280-17-23-001 SHEET: 01 OF 05 REV: 0

APPROVAL						
ENG MNGR	C & I	ELE	C OMM	ENG MNGR	ENG MNGR	ENG MNGR

SHEET SIZE : A0

CAD NO: 30201_50150_LT_HE-H4-280-17-23-001-0.dwg

NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS
IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER



FOUNDATION LAYOUT

* – 50mm ELASTIC TYPE JOINT SEALANT WITH FILLER BOARD. FLEXCELL OR SIMILLAR APPROVED.

ALL LOCATIONS AND SIZES OF OPENINGS/ SLEEVES/ PLINTH SHALL BE CHECKED WITH LATEST APPROVED ELECTRICAL, HVAC, FIRE FIGHTING, CO-ORDINATION DRAWINGS AND OTHER SERVICES DRAWING BEFORE CONCRETING

– FOUNDATIONS IDENTIFIED FOR CMS. TO BE VALIDATED BY CMS VENDOR

REFERENCE DRAWINGS

30201_50150_LT_HE-H4-262-13-60-001	UNIT PLOT PLAN – UTILITY AREA HABSHAN-4 (U261, U262 ,U264, U266, U267, U268, U272, U280)
30201_50150_LT_HE-H4-280-20-22-001	UNIT 213 – HABSHAN 4 ARCHITECTURAL LAYOUT AND DETAILS SUBSTATION SS-201
30201_50150_LT_HE-GG-000-17-11-002	TYPICAL CONCRETE GA AND REINFORCEMENT DETAILS
30201_50150_LT_HE-GG-000-16-21-001	TYPICAL INSTALLATION DETAILS (EARTHING AND LIGHTING SYSTEM)
30201_50150_LT_HE-H4-262-14-21-001	CIVIL GENERAL ARRANGEMENT AND UNDERGROUND COMPOSITE LAYOUT_HABSHAN 4 UTILITY
30201_50150_LT_HE-H4-280-17-23-002	STRUCTURAL CONCRETE LAYOUT AND DETAILS FOR SUBSTATION SS-201
30201_50150_LT_HE-GG-000-16-21-001	TYPICAL INSULATION DETAILS (EARTHING & LIGHTNING SYSTEM)
30201_50150_LT_HE-GG-000-17-31-002	DETAILS OF PROTECTION TO CONCRETE FOUNDATIONS AND PEDESTALS

FOUNDATION CONCRETE = 1085 CUM.

NOTE : ALL FOUNDATION QUANTITIES INCLUDED. FOR ALL WALL AND COLUMN QUANTITIES SHALL BE REFERED TO RESPECTIVE DRAWINGS.

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE.
2. ALL COORDINATES AND ELEVATIONS ARE IN METERS UNLESS NOTED OTHERWISE.
3. THE COORDINATES MENTIONED ARE IN PLANT COORDINATES AND ELEVATIONS MENTIONED ARE IN JEBEL DHANNA CHART DATUM (JDCD)
4. PLANT CO-ORDINATES E. 9431.910, N. 7903.650 EQUAL TO UTM CO-ORDINATES E.769200.960, N.2637528.035.
5. FOR GENERAL NOTES, ABBREVIATIONS AND SYMBOLS – CONCRETE WORKS REFER STANDARD DRAWING NO. 30201_50150_LT_HE-GG-000-17-11-001.

FOLLOWING ARE MINIMUM REQUIREMENTS FOR ALL CONCRETE WORKS:

MATERIAL OF CONSTRUCTION	
PARTICULARS	TYPE
BLINDING CONCRETE	f'c = 20MPa 28 DAY CUBE STRENGTH
RCC	f'c = 40MPa 28 DAY CUBE STRENGTH
REINFORCEMENT	fy = 500 N/MM2 (MINIMUM YEILD STRENGTH 500 MPA – BS44449)

CLEAR COVER TO REINFORCEMENT SHALL BE AS BELOW:			
TYPE	TOP	BOTTOM	SIDE
FOUNDATION / WALL / COLUMN BELOW GRADE	75	75	75
GRADE BEAM	75	75	75
COLUMN ABOVE GRADE	50	50	50

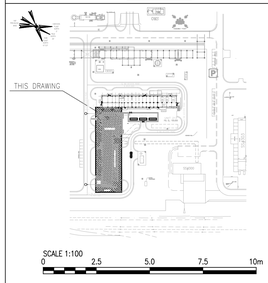
6. FOR DETAILS OF PROTECTION TO CONCRETE FOUNDATIONS AND PEDESTALS REFER STANDARD DRAWING 30201_50150_LT_HE-GG-000-17-31-002.
7. THIS FOUNDATION IS CONSIDERED TO BE CRITICAL FOUNDATION AND CORROSION MONITORING SYSTEM SHALL BE PROVIDED AS PER AGES STANDARD DRAWING AGES-STD-01-108. REBAR ELECTRICAL CONTINUITY SHALL BE CHECKED AND ENSURED BEFORE CONCRETE POURING.
8. DURING CONSTRUCTION ADEQUATE MEASURES SHALL BE TAKEN IF ACTUAL SITE CONDITION DIFFERS FROM THE EXISTING AS BUILT DRAWINGS AND MODELS.
9. ALL EXISTING FOUNDATIONS AND CONCRETE WORKS, TOGETHER WITH EXISTING UNDERGROUND SERVICES IN THE PROXIMITY OF THESE NEW WORKS SHALL BE PROTECTED FROM UNDERMINING AND DAMAGE DURING EXCAVATION, CONSTRUCTION AND BACKFILLING.
10. EPCM CONSTRUCTION CONTRACTOR SHALL ENSURE THAT EXCAVATION & EXECUTION SHALL BE CARRIED OUT DURING NORMAL WIND CONDITIONS ONLY.
11. EPCM CONSTRUCTION CONTRACTOR SHALL PERFORM SLIT TRENCH TO VERIFY THE PRESENCE OF EXISTING CABLES / UG LINES / OTHER UG FACILITIES ETC. BEFORE EXCAVATION FOR THE FOUNDATION WORKS. IF SAME ARE FOUND, SUCH EXISTING UG FACILITIES SHALL BE EXPOSED CAREFULLY AND PROTECTED SUITABLY DURING CONSTRUCTION OF FOUNDATION.
12. THE EXISTING FACILITIES SHALL BE DEMOLISHED AND REINSTATED AFTER THE CONSTRUCTION OF NEW FOUNDATIONS AND OTHER UTILITIES. DURING EXCAVATION, NECESSARY PROTECTION SHALL BE PROVIDED TO THE EMBANKMENT. THE EPC CONTRACTOR SHALL SUBMIT A METHOD STATEMENT TO THE COMPANY FOR APPROVAL BEFORE COMMENCING ANY CONSTRUCTION ACTIVITIES.
13. ALL CONCRETE WORKS SHALL BE CARRIED OUT AS PER ADDENDUM TO AGES-SP-01-001.
14. ALL EARTHWORKS SHALL BE CARRIED OUT AS PER AGES-SP-01-005.
15. GWT ELEVATION AS PER EXECUTE STAGE SOIL REPORT IS 71.110 (CONSIDERING 1M SEASONAL VARIATION) AS PER JDCD LEVEL.

FOUNDATION DESIGN ARE CARRIED OUT WITH 180 kN/M2 NET SBC AT 2.5m DEPTH AS PER EXECUTION STAGE GEOTECHNICAL REPORT No RPT-30201_50150_LT_HE-GG-999-14-00-222.

HOLD

1. BUILDING CO-ORDINATES ON HOLD
2. TRANSFORMER FIRE WALL LOCATIONS ARE ON HOLD.
3. COLUMN GRID DIMENSION AND LOCATION ON HOLD.
4. BATTERY ROOM ON HOLD.

KEY PLAN



REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
C	07-FEB-2025	ISSUED FOR CONSTRUCTION WITH HOLD (IFC-H)	RM	DR/KS	VK
B	30-JAN-2025	ISSUED FOR APPROVAL	KR	HU/KS	VK
A	17-JAN-2025	ISSUED FOR COMPANY REVIEW (ICR)	VA	AC/HB	VK/MS

SCALE:	DRAWN:	CHKD:	APPRD:
AS SHOWN	ASHOK	ANKITA/BALAJI	VISHWANATH/SOFI

EPCm CONTRACTOR: wood. EPCm CONTRACTOR: AMEC INTERNATIONAL LTD.

PROJECT TITLE: PROJECT 5.0 (P5) LONG TERM - EXECUTE STAGE EPCm FOR HABSHAN DBN

DRAWING TITLE: UNIT 280 –HABSHAN 4 FOUNDATION GA & DETAILS SUBSTATION SS-201 BUILDING FOUNDATION LAYOUT

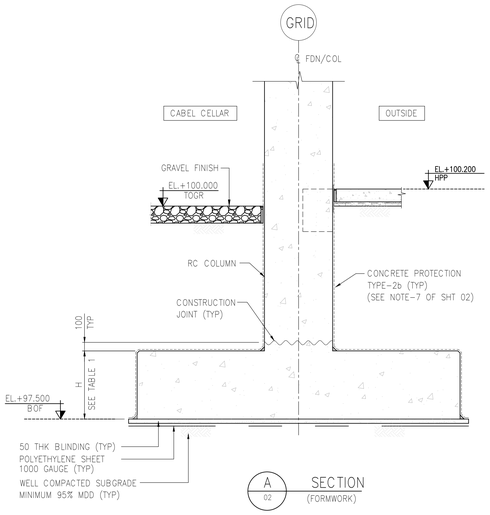
UNIT: 213	PLANT: H4
CONTR. PROJ. No. 3000076501	CONT. DWG. No.: XXXXXXXX
ADNOC GAS CONTRACT No. 3000076501	

ADNOC GAS PROJ. No.	ADNOC GAS DWG. No.	SHEET	REV.
30201_50150_LT_HE	30201_50150_LT_HE-H4-280-17-23-001	02 OF 05	0

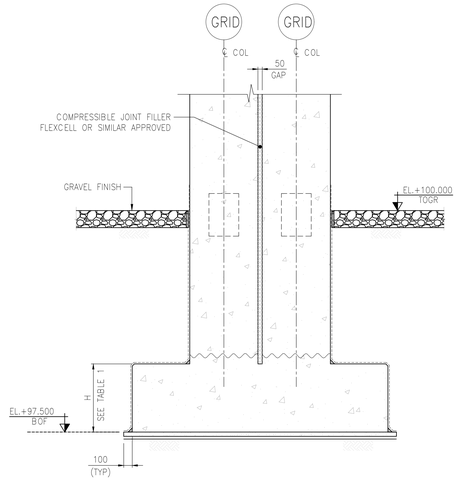
DWG NO: 30201_50150_LT_HE-H4-280-17-23-001-0.dwg

17 SHEET SIZE : A0

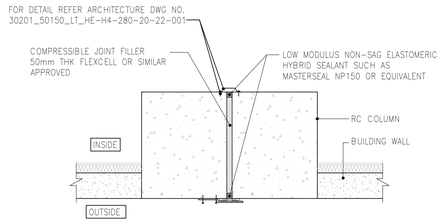
NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER



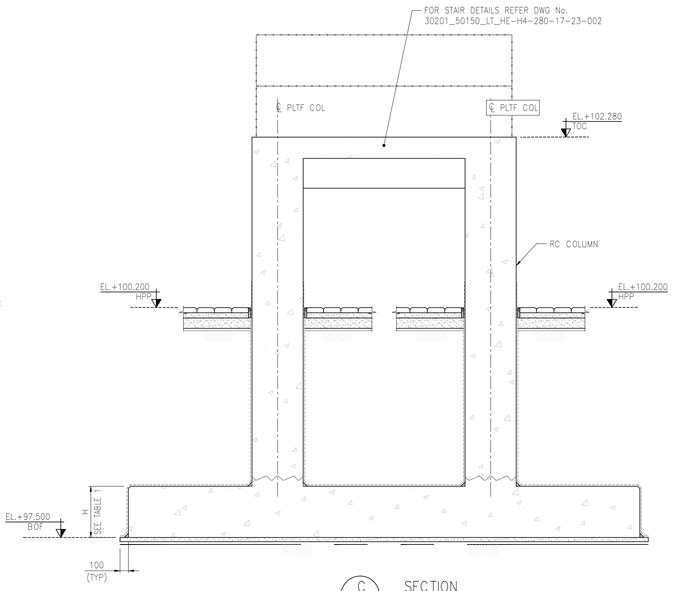
A 02 SECTION (FORMWORK)



B 02 SECTION (FORMWORK)



TYP EXPANSION JOINT AT COLUMNS



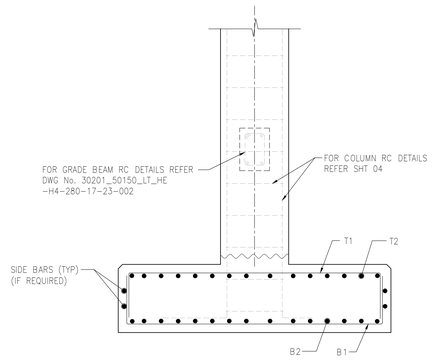
C 02 SECTION (FORMWORK)

NOTES

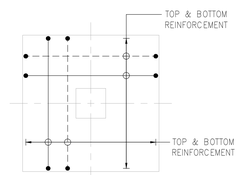
1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE.
2. ALL COORDINATES AND ELEVATIONS ARE IN METERS UNLESS NOTED OTHERWISE.
3. FOR ALL OTHER NOTES , KEY PLAN, REFERENCE DRAWINGS, REFER DWG NO. 30201_50150_LT_HE-H4-280-17-23-001 SHT 02.

LEGEND

℄	CENTERLINE
TYP	TYPICAL
HPP	HIGH POINT OF PAVING
FGL	FINISHED GRADE LEVEL
THK	THICKNESS
TOGR	TOP OF GRAVEL
FDN	FOUNDATION
COL/C1	COLUMN
BOF	BOTTOM OF FOUNDATION
TOC	TOP OF CONCRETE
--------	TOP BAR
———	BOTTOM BAR



TYPICAL REINFORCEMENT DETAILS
FOR FOOTING TYPES
CF1, CF2, CF3, CF4, CF5, CF6, CF7, F1, F2, F2a, F3, F3a & F4



TYPICAL PLAN FOOTING FOR – CF2, CF3, CF4, CF5, CF6, CF7, F1, F2, F2a, F3, F3a & F4
(SCALE 1:50)
(REINFORCEMENT)

TABLE-1

(FOR FOUNDATION ORIENTATION REFER SHT 02)

FOUNDATION MKD	QUANTITY	FOOTING DIMENSIONS		REINFORCEMENT				SIDE BARS	REMARKS
				AT TOP		AT BOTTOM			
		L x B	H	ALONG LENGTH (T1)	ALONG BREADTH (T2)	ALONG LENGTH (B1)	ALONG BREADTH (B2)		
CF1	1	(SEE LAYOUT)	800	T25-150	T20-150	T25-150	T20-150	4-T12 (EF)	
CF2	2	4500 x 2600	800	T20-150	T20-150	T20-150	T20-150	4-T12 (EF)	
CF3	2	4000 x 6000	600	T16-150	T16-150	T16-150	T16-150	2-T12 (EF)	
CF4	1	6000 x 6000	800	T20-150	T20-150	T20-150	T20-150	4-T12 (EF)	
CF5	4	4700 x 7300	800	T20-150	T20-150	T20-150	T20-150	4-T12 (EF)	
CF6	1	5500 x 7000	800	T20-150	T20-150	T20-150	T20-150	4-T12 (EF)	
CF7	2	6000 x 11500	900	T20-150	T20-150	T20-150	T20-150	4-T12 (EF)	
F1	16	4000 x 4000	800	T16-150	T16-150	T20-150	T20-150	4-T12 (EF)	
F1a	6	3500 x 3500	800	T16-150	T16-150	T20-150	T20-150	4-T12 (EF)	
F2	1	1900 x 1900	600	T16-150	T16-150	T20-150	T20-150	2-T12 (EF)	
F2a	1	2800 x 2800	800	T16-150	T16-150	T20-150	T20-150	4-T12 (EF)	
F3	8	3000 x 5650	600	T16-150	T16-150	T20-150	T20-150	2-T12 (EF)	
F3a	1	5050 x 5650	600	T16-150	T16-150	T20-150	T20-150	2-T12 (EF)	
F4	4	4700 x 4700	800	T16-150	T16-150	T20-150	T20-150	4-T12 (EF)	
F4a	7	4500 x 4500	800	T16-150	T16-150	T20-150	T20-150	4-T12 (EF)	

SCALE 1:25
0 0.5 1.0 1.5 2.0 2.5m

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
D	07-FEB-2026	ISSUED FOR CONSTRUCTION WITH HOLD (FCH)	RM	DR/KS	VV
B	30-JAN-2026	ISSUED FOR APPROVAL	KR	RG/KS	VV
A	17-JAN-2026	ISSUED FOR COMPANY REVIEW (ICR)	VA	AC/HB	VV/MS

SCALE: AS SHOWN | DRAWN: ASHOK | CHKD: ANKITA/BALAJI | APPRD: VISHWANATH/SOFI

ADNOC Gas

EPCm CONTRACTOR: wood. AMEC INTERNATIONAL LTD.

PROJECT TITLE: PROJECT 5.0 (P5)
LONG TERM - EXECUTE STAGE
EPCm FOR HABSHAN DBN

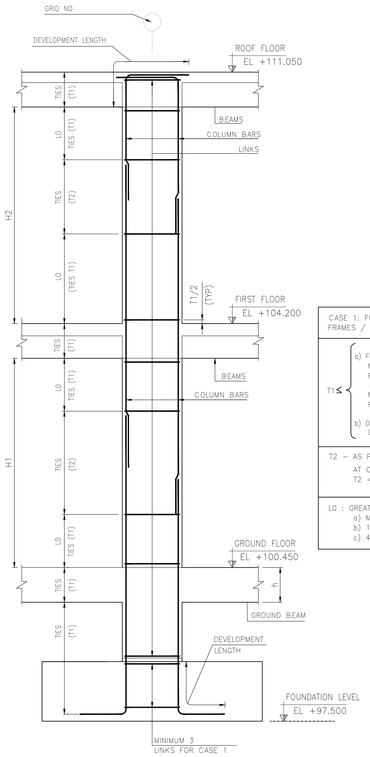
DRAWING TITLE: UNIT 280 –HABSHAN 4
FOUNDATION GA & DETAILS SUBSTATION SS-201
FOUNDATION SECTIONS & DETAILS

UNIT: 213 | PLANT: H4
CONTR. PROJ. No. 3000076501 | CONT. DWG. No.: XXXXXXXX
ADNOC GAS CONTRACT No. 3000076501
ADNOC GAS PROJ. No. 30201_50150_LT_HE | ADNOC GAS DWG. No. 30201_50150_LT_HE-H4-280-17-23-001 | SHEET 03 OF 05 | REV 0

APPROVAL						
ENG MNGR	C & I	ELE	C OMM	ENG MNGR	ENG MNGR	ENG MNGR

SIZE : A0

NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER



CASE 1: FOR INTERMEDIATE MOMENT FRAMES / SEISMIC DESIGN CATEGORY C

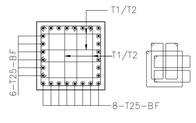
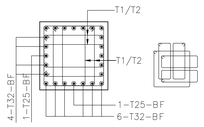
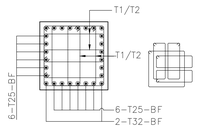
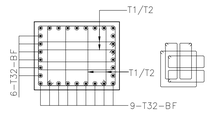
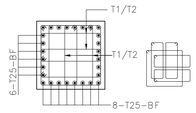
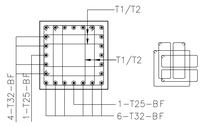
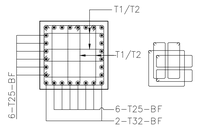
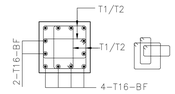
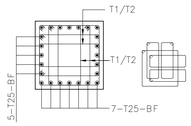
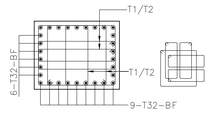
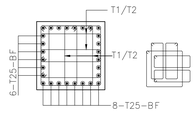
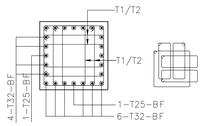
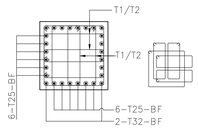
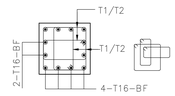
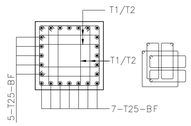
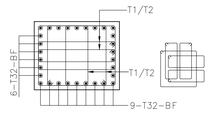
T1 ≤ { a) FOR GRADE 420, MIN OF (8 TIMES SMALLEST LONGITUDINAL REINFORCEMENT BAR DIAMETER, 200mm) FOR GRADE 550, MIN OF (6 TIMES SMALLEST LONGITUDINAL REINFORCEMENT BAR DIAMETER, 150mm); b) 0.5 SMALLEST CROSS-SECTIONAL DIMENSION OF THE COLUMN.

T2 – AS PER DESIGN AT COLUMN SPLICE LOCATION, T2 = MIN (d/4, 100mm), WHERE d IS COLUMN LEAST DIMENSION

L0 : GREATER OF THE FOLLOWING
a) MAX COLUMN DIMENSION
b) 1/6 OF CLEAR HEIGHT OF COLUMN
c) 450mm

TYPICAL COLUMN RC DETAIL
(SCALE: NTS)

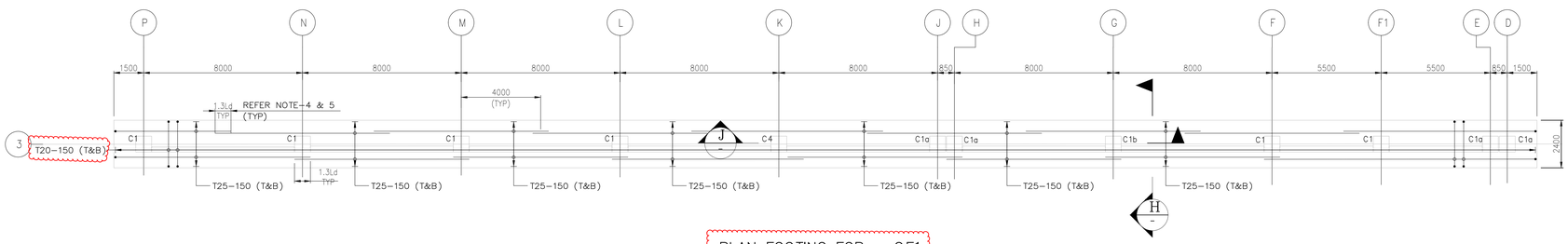
COLUMN REINFORCEMENT SCHEDULE

FLOOR \ COLUMN ID	C1 (800X800)		C1a (800X800)		C1b (800X800)		C2 & C2a (600X600)		C3 (800X800)		C4 (1000X800)	
	T1	T2	T1	T2	T1	T2	T1	T2	T1	T2	T1	T2
	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100	T10–100
FIRST FLOOR BEAM TO ROOF BEAM (EL +104.200 TO +111.050)												
GRADE BEAM TO FIRST FLOOR BEAM (EL +100.300 TO +102.200 FOR COLUMN C2a) (EL +100.300 TO +104.200 FOR COLUMNS C2 & C3)												
FOUNDATION TO GRADE BEAM (EL +97.500 TO +100.300)												

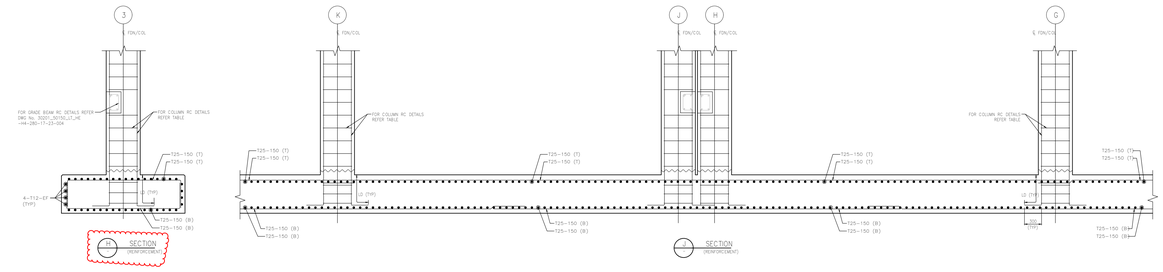
NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE.
2. ALL COORDINATES AND ELEVATIONS ARE IN METERS UNLESS NOTED OTHERWISE.
3. FOR ALL OTHER NOTES , KEY PLAN, REFERENCE DRAWINGS, REFER DWG NO. 30201_50150_LT_HE-H4-280-17-23-001 SHT 02.
4. DEVELOPMENT LENGTH LD SHALL BE MINIMUM 50xBAR DIA SPLICE LAPPING LENGTH SHALL BE 1.3 x LD. THE CENTER TO CENTER OF SPLICE STAGGERING SHALL NOT BE LESS THAN 1.3 LD.
5. TYPICAL SPLICE LOCATION PROVIDED BASED ON THAT SUITABLE SPLICE LOCATION SHALL BE ARRIVED DURING BAR BENDING SCHEDULE WHEREVER NOT SHOWN IN THE REINFORCEMENT PLAN.

50% OF REBAR SHALL BE SPLICED AT ONE LOCATION.



PLAN FOOTING FOR – CF1
(SCALE 1:100) (REINFORCEMENT)



SCALE 1:50
0 1.0 2.0 3.0 4.0 5.0m

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
0	07-FEB-2026	ISSUED FOR CONSTRUCTION WITH HOLD (IFCH)	RM	DR/KS	VV
B	30-JAN-2026	ISSUED FOR APPROVAL	KR	HG/KS	VV
A	17-JAN-2026	ISSUED FOR COMPANY REVIEW (CR)	VA	AC/HB	VV/MS

SCALE: AS SHOWN | DRAWN: ASHOK | CHKD: ANKITA/BALAJI | APPRD: VISHWANATH/SOFI

أدنوك للغاز ADNOC Gas

EPCm CONTRACTOR: wood. AMEC INTERNATIONAL LTD.

PROJECT TITLE: PROJECT 5.0 (P5)
LONG TERM - EXECUTE STAGE
EPCm FOR HABSHAN DBN

DRAWING TITLE: UNIT 280 –HABSHAN 4
FOUNDATION GA & DETAILS SUBSTATION SS–201
COLUMN SECTIONS AND RAFT FOUNDATION DETAILS

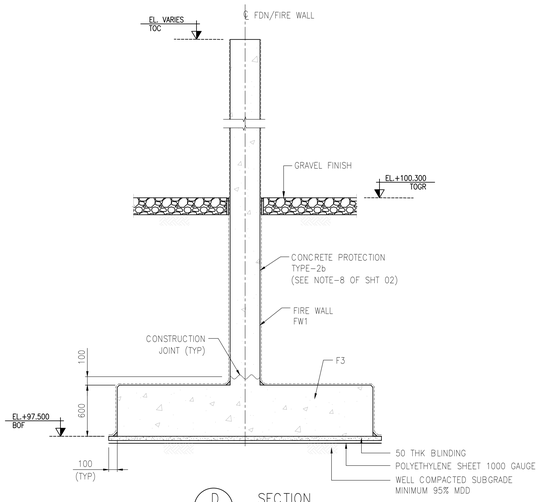
APPROVAL						
ENG MNGR	C & I	ELE	C OMM	ENG MNGR	ENG MNGR	ENG MNGR

UNIT: 213 | PLANT: H4
CONTR. PROJ. No. 3000076501 | CONT. DWG. No.: XXXXXXXX
ADNOC GAS CONTRACT No. 3000076501
ADNOC GAS PROJ. No.: 30201_50150_LT_HE | ADNOC GAS DWG. No.: 30201_50150_LT_HE-H4-280-17-23-001 | SHEET: 04 OF 05 | REV: 0

CAD NO: 30201_50150_LT_HE-H4-280-17-23-001-01.dwg

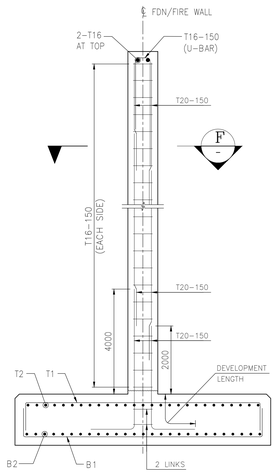
NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER

SHEET SIZE : A0

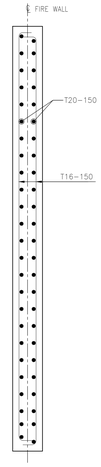


D 02 SECTION
(SCALE 1:25)
(FORMWORK)

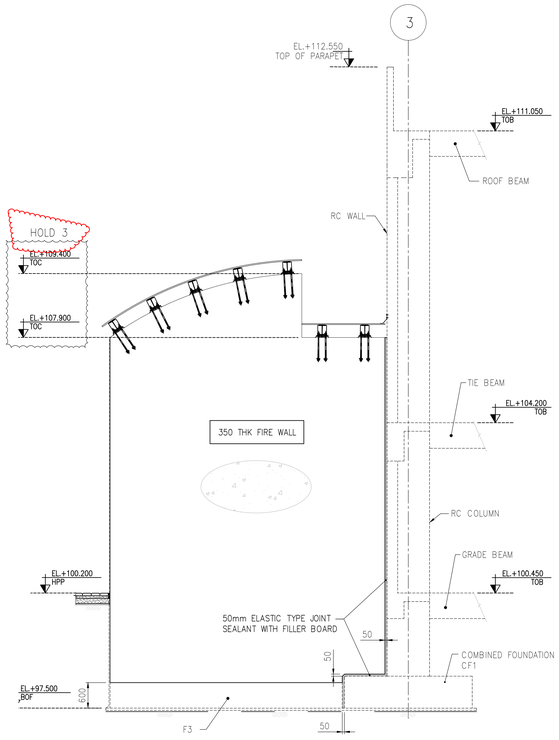
TYPICAL FIRE WALL SECTION



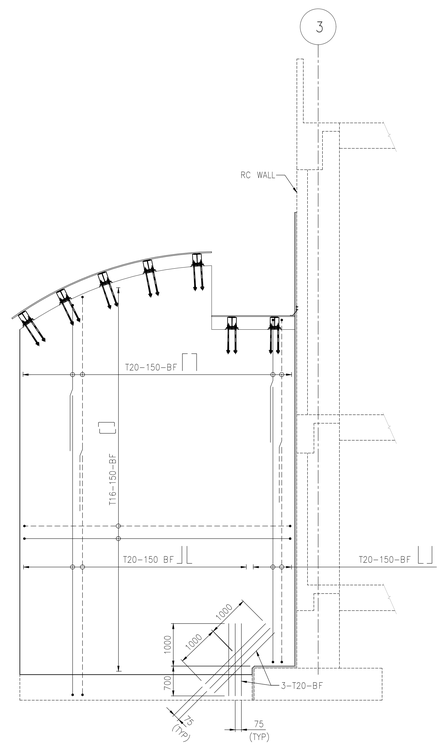
D 02 SECTION
(SCALE 1:25)
(REINFORCEMENT)



F - SECTION
(SCALE 1:25)
(REINFORCEMENT)



E 02 SECTION
(SCALE 1:50)
(FORMWORK)



E 02 SECTION
(SCALE 1:50)
(REINFORCEMENT)

NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE.
2. ALL COORDINATES AND ELEVATIONS ARE IN METERS UNLESS NOTED OTHERWISE.
3. FOR ALL OTHER NOTES , KEY PLAN, REFERENCE DRAWINGS, REFER DWG NO. 30201_50150_LT_HE-H4-280-17-23-001 SHT 02.

HOLD

3. HOLD-3 LEVEL ON HOLD.

SCALE 1:50
0 1.0 2.0 3.0 4.0 5.0m

REV	DATE	DESCRIPTION	DRAWN	CHKD	APPRD
0	07-FEB-2026	ISSUED FOR CONSTRUCTION WITH HOLD (IFCH)	RM	DR/KS	VV
B	30-JAN-2026	ISSUED FOR APPROVAL	KR	HG/KS	VV
A	17-JAN-2026	ISSUED FOR COMPANY REVIEW (CR)	VA	AC/HB	VV/MS

SCALE: AS SHOWN
DRAWN: ASHOK
CHKD: ANKITA/BALAJI
APPRD: VISHWANATH/SOFI

ADNOC Gas

EPCm CONTRACTOR: wood.
EPCm CONTRACTOR: AMEC INTERNATIONAL LTD.

PROJECT TITLE
PROJECT 5.0 (P5)
LONG TERM - EXECUTE STAGE
EPCm FOR HABSHAN DBN

DRAWING TITLE
UNIT 280 –HABSHAN 4
FOUNDATION GA & DETAILS SUBSTATION SS-201
WALL SECTIONS & DETAILS

APPROVAL						
ENG MNGR	C & I	ELE	C OMM	ENG MNGR	ENG MNGR	ENG MNGR

UNIT: 213 | PLANT: H4
CONTR. PROJ. No. 3000076501 | CONT. DWG. No.: XXXXXXXX
ADNOC GAS CONTRACT No. 3000076501
ADNOC GAS PROJ. No. 30201_50150_LT_HE | ADNOC GAS DWG. No. 30201_50150_LT_HE-H4-280-17-23-001 | SHEET 05 OF 05 | REV 0
CAD NO: 30201_50150_LT_HE-H4-280-17-23-001-Layg
SHEET SIZE : A0

NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER